

Why GROUP BY Exists

Suppose the table is:

id	department	salary
1	IT	50000
2	IT	60000
3	HR	40000
4	HR	45000
5	Sales	55000

Question:

Average salary of each department

Without GROUP BY, SQL doesn't know where one department ends and another begins.

GROUP BY tells SQL:

- Put all rows having the same department together.

Groups formed:

IT	HR	Sales
50000	40000	55000
60000	45000	

Now SQL calculates averages separately.

```
SELECT department,  
       AVG(salary)  
FROM Employee  
GROUP BY department;
```

Output

department	AVG(salary)
IT	55000
HR	42500
Sales	55000

Row-Level

Every input row produces one output row.

Example

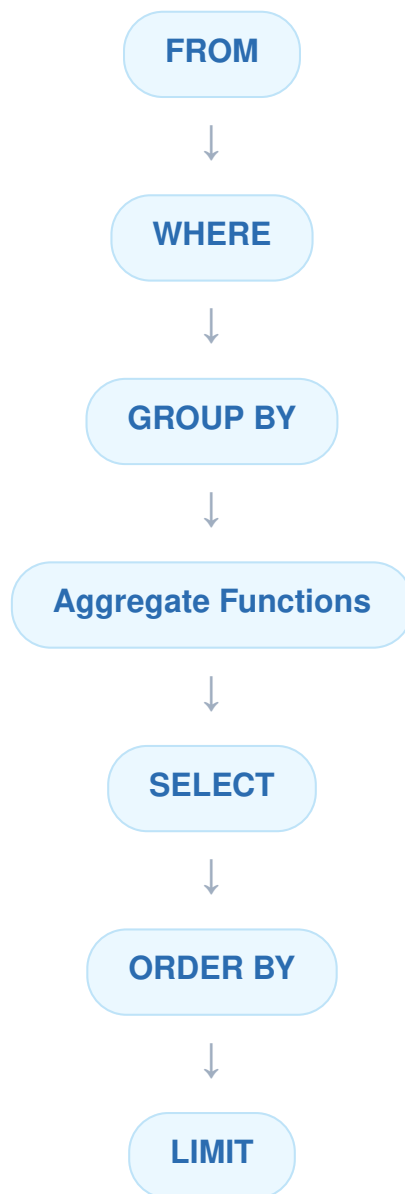
```
SELECT name,  
       salary * 1.1  
FROM Employee;
```

Group-Level

Many rows become one row.

```
SELECT department,  
       AVG(salary)  
FROM Employee  
GROUP BY department;
```

SQL executes as



COUNT() with GROUP BY

COUNT() becomes much more useful when combined with GROUP BY.

14.2.1 COUNT(*)

Counts all rows in each group.

Example

department
IT
IT
HR
HR
Sales

```
SELECT department,
       COUNT(*)
FROM Employee
GROUP BY department;
```

Output

department	COUNT(*)
IT	2
HR	2
Sales	1

14.2.2 COUNT(column)

Counts only non-NULL values.

Example

department	phone
IT	111
IT	NULL
IT	222

```
SELECT department,
       COUNT(phone)
FROM Employee
GROUP BY department;
```

Output

department	COUNT(phone)
IT	2

The NULL phone is ignored.

14.2.3 COUNT(DISTINCT column)

Counts unique values inside each group.

Example

department	city
IT	Delhi
IT	Delhi
IT	Mumbai
IT	Delhi

```
SELECT department,  
       COUNT(DISTINCT city)  
FROM Employee  
GROUP BY department;
```

Output

department	COUNT(DISTINCT city)
IT	2

Only Delhi and Mumbai.

14.2.4 NULL Handling

Suppose

department	bonus
IT	1000
IT	NULL
IT	2000

COUNT(*)	Result	3
COUNT(bonus)	Result	2

GROUP BY Multiple Columns

Sometimes one column isn't enough.

Example

department	city	salary
IT	Delhi	50000
IT	Delhi	60000
IT	Mumbai	70000
HR	Delhi	40000
HR	Mumbai	45000

Group by Department

```
GROUP BY department
```

Creates

- IT
- HR

Group by Department and City

```
GROUP BY department, city
```

Creates

- IT + Delhi
- IT + Mumbai
- HR + Delhi
- HR + Mumbai

Each unique combination becomes a separate group.

Example

```
SELECT department,  
       city,  
       COUNT(*) AS employees  
FROM Employee  
GROUP BY department, city;
```

Group by Year and Month

```
SELECT YEAR(order_date),  
       MONTH(order_date),  
       COUNT(*)  
FROM Orders  
GROUP BY YEAR(order_date),  
         MONTH(order_date);
```

Useful for monthly reports.

Question

Count employees in salary ranges.

- Low
- Medium
- High

Query


```
SELECT
    CASE
        WHEN salary < 40000 THEN 'Low'
        WHEN salary < 70000 THEN 'Medium'
        ELSE 'High'
    END AS salary_range,
    COUNT(*)
FROM Employee
GROUP BY
    CASE
        WHEN salary < 40000 THEN 'Low'
        WHEN salary < 70000 THEN 'Medium'
        ELSE 'High'
    END;
```

GROUP BY CONCAT()

Suppose

first_name	last_name
John	Smith
John	Smith
Alice	Brown

```
SELECT
    CONCAT(first_name, ' ', last_name) AS full_name,
    COUNT(*)
FROM Employee
GROUP BY CONCAT(first_name, ' ', last_name);
```

Output

full_name	count
John Smith	2
Alice Brown	1

GROUP BY EXTRACT()

PostgreSQL

```
SELECT
    EXTRACT(YEAR FROM order_date),
    COUNT(*)
FROM Orders
GROUP BY EXTRACT(YEAR FROM order_date);
```